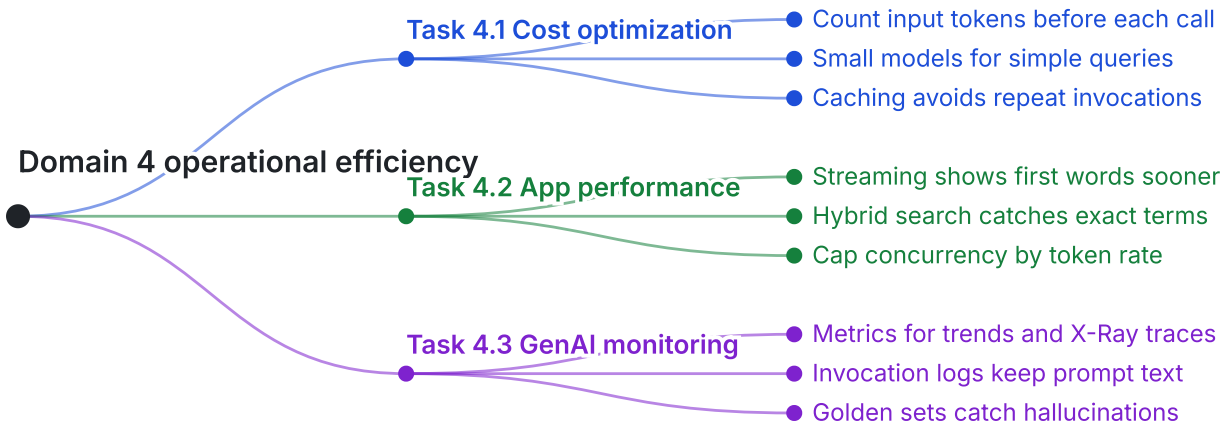
Memory sheet: Domain 4, operational efficiency

Exam weight 12 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
Steady, all-day, or reserved load	Provisioned Throughput	Spiky load, which pays for idle hours
Spike, sale, or unpredictable traffic	On-demand calls with token budgets	A steady load that costs less when reserved
Overnight, backlog, or by morning	Batch inference from S3	A chat screen waiting for each answer
Typing, first words, or a slow start	Response streaming	A batch job with no user waiting
Same question, repeated, or nightly refresh	Caching or precomputed answers	Personal answers, which must not be shared
Exact code, product number, or rare term	Hybrid search	A broad meaning question with no exact term
Spend jump across services or accounts	Cost Anomaly Detection	A runtime burst that needs an alarm now, since cost data can lag
Trend, spike, p50, or threshold on a metric	CloudWatch metrics and alarms	One slow request, where X-Ray shows the path

Etch lines

1. Measure the prompt size before calling, and set an output ceiling.
2. Average latency hides slow outliers, so check p50 and p90 too.
3. TimeToFirstToken covers streams, while OTPS isolates model throughput.
4. Steady or reserved wording signals Provisioned Throughput.

5. Throttled calls are missing from Invocations, so check the throttle metric.
6. Prompt text lives in invocation logs, while X-Ray covers timing.

Flash questions

- Q: Which capacity model fits a steady load on one model all day? A: Provisioned Throughput.
- Q: What does a token-burst alarm watch for each model? A: Sudden spikes in input or output token counts against a learned band.
- Q: Which streaming metric shows the wait before the first token arrives? A: TimeToFirstToken.
- Q: Which trace tool shows where time went across API Gateway, Lambda, and a model call? A: AWS X-Ray.
- Q: How far can Cost Anomaly Detection data lag behind actual spend? A: Up to 24 hours.
- Q: Which metric counts throttled model requests? A: InvocationThrottles.
- Q: Name one kind of answer that should stay out of a shared cache. A: A personal answer.
- Q: Which steps cut input tokens before each model call? A: Pruning history and chunks, plus compressing repeated instructions.

Source: domain4.md, Tasks 4.1 to 4.3, and frags2 t4.1.html to t4.3.html. AWS docs and the study guide sections above, no practice-test text.